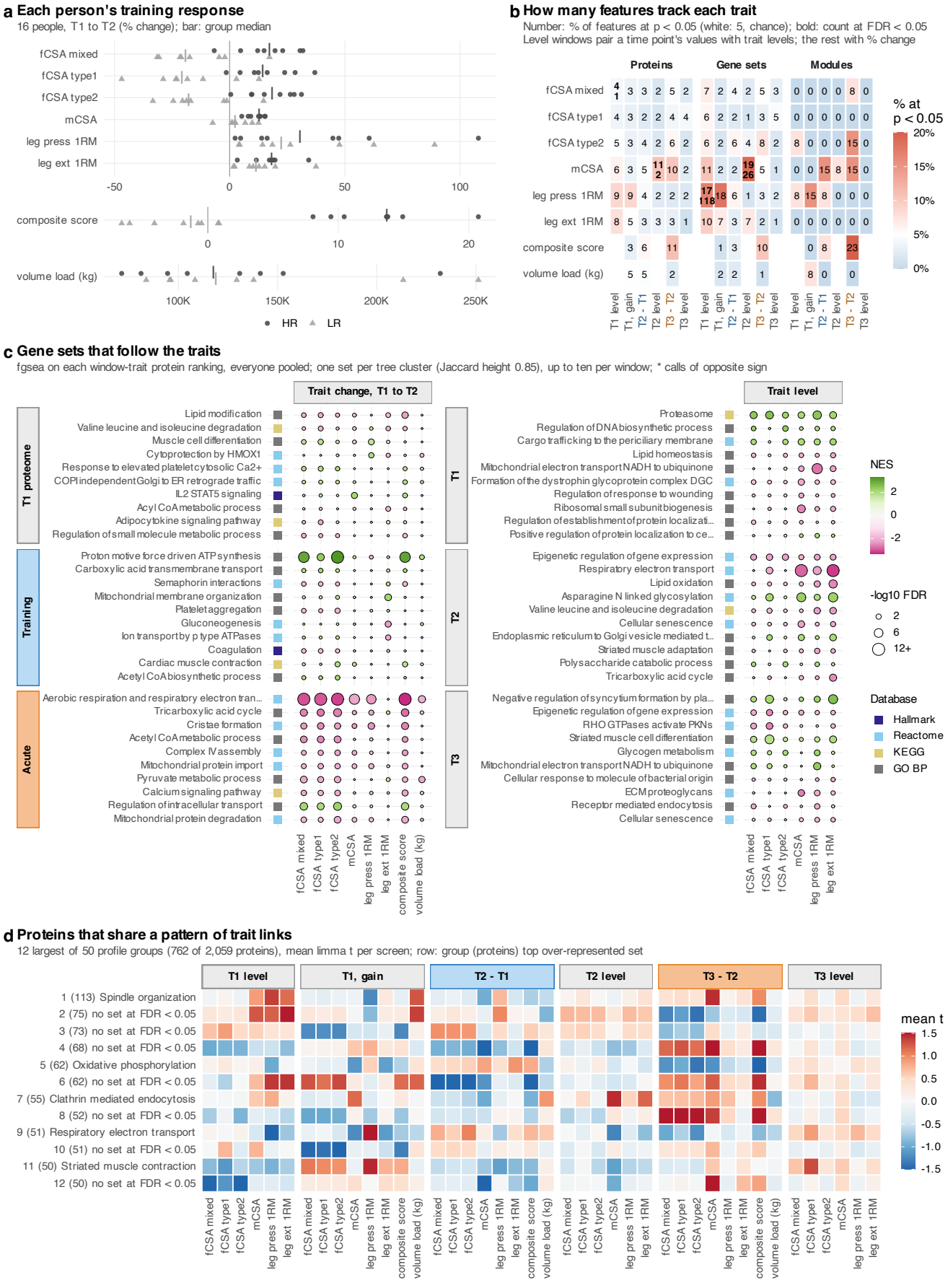

a Each person's training response
16 people, T1 to T2 (% change); bar: group median
fCSA mixed
fCSA type1
fCSA type2
mCSA
leg press 1RM
leg ext 1RM
composite score
volume load (kg)
HR
LR
b How many features track each trait
Number: % of features at p < 0.05 (white: 5, chance); bold: count at FDR < 0.05
Level windows pair a time point's values with trait levels; the rest with % change
Proteins
Gene sets
Modules
% at p < 0.05
T1 level
T1, gain
T2 - T1
T2 level
T3 - T2
T3 level
c Gene sets that follow the traits
fgsea on each window-trait protein ranking, everyone pooled; one set per tree cluster (Jaccard height 0.85), up to ten per window; * calls of opposite sign
Trait change, T1 to T2
Trait level
T1 proteome
Training
Acute
T1
T2
T3
Lipid modification
Valine leucine and isoleucine degradation
Muscle cell differentiation
Cytoprotection by HMOX1
Response to elevated platelet cytosolic Ca2+
COPI independent Golgi to ER retrograde traffic
IL2 STAT5 signaling
Acyl CoA metabolic process
Adipocytokine signaling pathway
Regulation of small molecule metabolic process
Proton motive force driven ATP synthesis
Carboxylic acid transmembrane transport
Semaphorin interactions
Mitochondrial membrane organization
Platelet aggregation
Gluconeogenesis
Ion transport by p type ATPases
Coagulation
Cardiac muscle contraction
Acetyl CoA biosynthetic process
Aerobic respiration and respiratory electron tran...
Tricarboxylic acid cycle
Cristae formation
Acetyl CoA metabolic process
Complex IV assembly
Mitochondrial protein import
Pyruvate metabolic process
Calcium signaling pathway
Regulation of intracellular transport
Mitochondrial protein degradation
Proteasome
Regulation of DNA biosynthetic process
Cargo trafficking to the periciliary membrane
Lipid homeostasis
Mitochondrial electron transport NADH to ubiquinone
Formation of the dystrophin glycoprotein complex DGC
Regulation of response to wounding
Ribosomal small subunit biogenesis
Regulation of establishment of protein localizati...
Positive regulation of protein localization to ce...
Epigenetic regulation of gene expression
Respiratory electron transport
Lipid oxidation
Asparagine N linked glycosylation
Valine leucine and isoleucine degradation
Cellular senescence
Endoplasmic reticulum to Golgi vesicle mediated t...
Striated muscle adaptation
Polysaccharide catabolic process
Tricarboxylic acid cycle
Negative regulation of syncytium formation by pla...
Epigenetic regulation of gene expression
RHO GTPases activate PKNs
Striated muscle cell differentiation
Glycogen metabolism
Mitochondrial electron transport NADH to ubiquinone
Cellular response to molecule of bacterial origin
ECM proteoglycans
Receptor mediated endocytosis
Cellular senescence
NES
-log10 FDR
Database
Hallmark
Reactome
KEGG
GO BP
d Proteins that share a pattern of trait links
12 largest of 50 profile groups (762 of 2,059 proteins), mean limma t per screen; row: group (proteins) top over-represented set
1 (113) Spindle organization
2 (75) no set at FDR < 0.05
3 (73) no set at FDR < 0.05
4 (68) no set at FDR < 0.05
5 (62) Oxidative phosphorylation
6 (62) no set at FDR < 0.05
7 (55) Clathrin mediated endocytosis
8 (52) no set at FDR < 0.05
9 (51) Respiratory electron transport
10 (51) no set at FDR < 0.05
11 (50) Striated muscle contraction
12 (50) no set at FDR < 0.05
mean t